01. Scope & addressing

Enterprise hierarchical network | Duncar Network | Revised October 29, 2026

Purpose: rebuild and verify the existing Packet Tracer lab using VLANs, trunks, Rapid PVST+, SVIs, OSPF, centralized DHCP, a default route and PAT. This is a lab procedure, not a production change plan.

VLAN / role	Subnet	Core-SW1 / Core-SW2
10 / Staff	10.10.0.0/23	10.10.0.1 / 10.10.0.2
20 / IT	10.20.0.0/24	10.20.0.1 / 10.20.0.2
30 / Admin	10.30.0.0/24	10.30.0.1 / 10.30.0.2
90 / Servers	10.90.0.0/24	10.90.0.1 / 10.90.0.2
99 / Management	10.99.0.0/24	10.99.0.1 / 10.99.0.2
SW1 - R1	10.255.255.0/30	.1 / .2
SW2 - R1	10.255.255.4/30	.5 / .6
R1 - ISP	203.0.113.0/30	.1 / .2

Before you start

Save a separate pre-change .pkt copy. Capture show running-config, show ip interface brief and show interfaces trunk where supported. Use actual connected ports; angle-bracket names below are placeholders, not commands to paste.

Design limit: client gateways use Core-SW1 (.1). Core-SW2 (.2) does not automatically take over those addresses. No HSRP is configured in the source MOP. Redundant links do not provide complete gateway redundancy.

02. Device baseline & VLANs

Apply to core, distribution and access switches. Set a unique hostname on each device. The passwords below preserve the original lab; they are not production credentials. VTY password login is not an SSH configuration, and service password-encryption is weak obfuscation.

```
enable
configure terminal
hostname &lt;unique-device-name&gt;
enable secret class
service password-encryption
line console 0
 password cisco
 login
 exit
line vty 0 4
 password cisco
 login
 exit
lldp run
vlan 10
 name Staff
vlan 20
 name IT
vlan 30
 name Admin
vlan 90
 name Servers
vlan 99
 name Management
end
```

Verify: show vlan brief and show lldp neighbors. LLDP availability varies by device model.

03. Access ports & trunks

Endpoint-facing access ports

Example: two Staff PCs on Fa0/1-2. Repeat on each access switch using the VLAN actually assigned to its endpoints. Do not paste all VLAN examples onto the same ports: the last assignment replaces the earlier one.

```
configure terminal
interface range fa0/1 - 2
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
end
```

Use VLAN 20 for IT, 30 for Admin and 90 for server endpoints. Your screenshot and original switch-order notes differ: follow the intended endpoint role and verify actual assignments. PortFast belongs on endpoint ports, not ordinary inter-switch links.

Inter-switch uplinks

```
configure terminal
interface range &lt;actual-uplink-ports&gt;
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,90,99
 no shutdown
end
```

On models that require it, enter switchport trunk encapsulation dot1q before switchport mode trunk. Omit that command on fixed-802.1Q models such as the 2960. Router-facing core ports are routed ports, not trunks.

Verify: show interfaces trunk; show vlan brief; show interfaces fa0/1 switchport.

04. Spanning tree & core interconnect

Core-SW1

```
configure terminal
spanning-tree mode rapid-pvst
spanning-tree vlan 10,30,90 root primary
spanning-tree vlan 20,99 root secondary
end
```

Core-SW2

```
configure terminal
spanning-tree mode rapid-pvst
spanning-tree vlan 20,99 root primary
spanning-tree vlan 10,30,90 root secondary
end
```

On every other switch: configure spanning-tree mode rapid-pvst. The source design uses independent trunks between the cores, not EtherChannel. STP may place a redundant link in an alternate/discarding state.

Only if removing the old Port-channel10

Confirm membership first with show etherchannel summary. On SW1 the source lists Gi1/0/22-23; on SW2 Gi1/0/23-24. Verify cabling before using these ranges.

```
configure terminal
interface range &lt;confirmed-old-channel-members&gt;
 no channel-group 10
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,90,99
 no shutdown
exit
no interface port-channel 10
end
```

Verify: show spanning-tree vlan 10; show spanning-tree vlan 20; show interfaces trunk. Confirm the intended root bridge on each VLAN.

05. SVIs & server addressing

On both core multilayer switches, enable ip routing. Configure every SVI using the table on page 1. The full pattern below is for Core-SW1; on Core-SW2 use .2 instead of .1 for each VLAN.

```
configure terminal
ip routing
interface vlan 10
 ip address 10.10.0.1 255.255.254.0
 no shutdown
interface vlan 20
 ip address 10.20.0.1 255.255.255.0
 no shutdown
interface vlan 30
 ip address 10.30.0.1 255.255.255.0
 no shutdown
interface vlan 90
 ip address 10.90.0.1 255.255.255.0
 no shutdown
interface vlan 99
 ip address 10.99.0.1 255.255.255.0
 no shutdown
end
```

Server0

Desktop > IP Configuration: IP 10.90.0.10, mask 255.255.255.0, gateway 10.90.0.1. On its connected switch port:

```
configure terminal
interface &lt;server-facing-port&gt;
 switchport mode access
 switchport access vlan 90
 spanning-tree portfast
end
```

Verify: show ip interface brief; show ip route. SVIs need an active VLAN and a participating forwarding port to become operational. Test a client gateway and cross-VLAN ping.

06. Routed uplinks & OSPF

Routed link configuration

```
! Core-SW1
configure terminal
interface gi1/0/20
 no switchport
 ip address 10.255.255.1 255.255.255.252
 no shutdown
end
! Core-SW2
configure terminal
interface gi1/0/20
 no switchport
 ip address 10.255.255.5 255.255.255.252
 no shutdown
end
! Core-R1
configure terminal
interface gi8/0
 ip address 10.255.255.2 255.255.255.252
 no shutdown
interface gi9/0
 ip address 10.255.255.6 255.255.255.252
 no shutdown
router ospf 1
 router-id 1.1.1.1
 network 10.255.255.0 0.0.0.3 area 0
 network 10.255.255.4 0.0.0.3 area 0
end
```

The /30 subnets are distinct: .0 has usable .1-.2; .4 has usable .5-.6. Their broadcast addresses are .3 and .7.

Verify: show ip interface brief. Ping each directly connected peer before checking OSPF.

07. OSPF on the core switches

Core-SW1

```
configure terminal
router ospf 1
 network 10.255.255.0 0.0.0.3 area 0
 network 10.10.0.0 0.0.1.255 area 0
 network 10.20.0.0 0.0.0.255 area 0
 network 10.30.0.0 0.0.0.255 area 0
 network 10.90.0.0 0.0.0.255 area 0
 network 10.99.0.0 0.0.0.255 area 0
end
```

Core-SW2

```
configure terminal
router ospf 1
 network 10.255.255.4 0.0.0.3 area 0
 network 10.10.0.0 0.0.1.255 area 0
 network 10.20.0.0 0.0.0.255 area 0
 network 10.30.0.0 0.0.0.255 area 0
 network 10.90.0.0 0.0.0.255 area 0
 network 10.99.0.0 0.0.0.255 area 0
end
```

Verify: show ip ospf neighbor; show ip route ospf; show ip protocols. Expect FULL adjacency on each core-to-router routed link. These statements also enable OSPF on matching SVIs; additional core-to-core adjacencies may form across the shared VLANs.

Teaching point: 10.10.0.0/23 uses wildcard 0.0.1.255. The /24 VLANs use 0.0.0.255. OSPF network statements select matching local interfaces.

08. DHCP pools & relay

Core-R1: address pools

```
configure terminal
ip dhcp excluded-address 10.10.0.1 10.10.0.2
ip dhcp excluded-address 10.20.0.1 10.20.0.2
ip dhcp excluded-address 10.30.0.1 10.30.0.2
ip dhcp pool STAFF
 network 10.10.0.0 255.255.254.0
 default-router 10.10.0.1
 dns-server 8.8.8.8
exit
ip dhcp pool IT
 network 10.20.0.0 255.255.255.0
 default-router 10.20.0.1
 dns-server 8.8.8.8
exit
ip dhcp pool ADMIN
 network 10.30.0.0 255.255.255.0
 default-router 10.30.0.1
 dns-server 8.8.8.8
end
```

Core switches: relay on client SVIs

On Core-SW1, apply the following to VLANs 10, 20 and 30, changing the interface number each time. On Core-SW2, use helper address 10.255.255.6 instead.

```
configure terminal
interface vlan 10
 ip helper-address 10.255.255.2
end
```

Verify: Set each client to DHCP. Run ipconfig /all on the PC, and show ip dhcp pool / show ip dhcp binding on Core-R1. No server or management DHCP pools are defined here. The DNS option does not create a reachable DNS service in Packet Tracer.

09. WAN & cloud mapping

Router WAN interfaces

```
! Core-R1
configure terminal
interface fa0/0
 ip address 203.0.113.1 255.255.255.252
 no shutdown
end
! INTERNET ISR4331
configure terminal
interface gi0/0/0
 ip address 203.0.113.2 255.255.255.252
 no shutdown
end
```

Cloud DSL mapping - confirmed lab fix

Open Cloud0 > Config > Connections > DSL. Select Modem9 on the left and GigabitEthernet4 on the right, then click Add. Confirm the row appears. These are the ports observed in this lab; use the actual connected ports if rebuilding with different cabling.

Path: Core-R1 Fa0/0 connects to cloud GigabitEthernet4; cloud Modem9 connects to the DSL modem; modem Ethernet connects to INTERNET Gi0/0/0.

Verify: On Core-R1: ping 203.0.113.2, then show arp. On INTERNET: ping 203.0.113.1. Retry the first ping to allow ARP resolution.

Troubleshooting lesson

Observed: both WAN interfaces were up/up, pings failed, and Core-R1 had no ARP entry for 203.0.113.2. The cloud DSL mapping table was empty. Adding Modem9 to GigabitEthernet4 restored the router-to-ISP ping. Up/up alone did not prove end-to-end Layer 2 forwarding.

These addresses represent a simulated ISP connection. They do not provide real internet access.

10. Default route & PAT

Core-R1

```
configure terminal
ip route 0.0.0.0 0.0.0.0 203.0.113.2
router ospf 1
 default-information originate
exit
access-list 1 permit 10.0.0.0 0.255.255.255
ip nat inside source list 1 interface fa0/0 overload
interface gi8/0
 ip nat inside
interface gi9/0
 ip nat inside
interface fa0/0
 ip nat outside
end
```

The static default route points unknown destinations toward the ISP. OSPF advertises that default when it is present in the routing table. PAT translates matching inside traffic to the Fa0/0 address; ACL 1 selects addresses for translation, rather than acting as an interface filtering policy.

Verification sequence

```
! Core switches
show ip route
! Look for O*E2 0.0.0.0/0

! Staff PC: generate traffic
ping 203.0.113.2

! Core-R1: inspect promptly before translations expire
show ip nat translations
show ip nat statistics
show access-lists
```

Verify: Client ping succeeds and the translation table shows its private source translated to 203.0.113.1. Client-to-ISP ping was confirmed working after the DSL mapping fix.

11. Acceptance, recording & recovery

Final acceptance checklist

[ ] PC has the correct DHCP address, mask and .1 gateway.

[ ] PC can ping its gateway and Server0 at 10.90.0.10.

[ ] Trunks carry VLANs 10,20,30,90,99; access ports match endpoints.

[ ] Core-SW1 is STP root for 10,30,90; Core-SW2 for 20,99.

[ ] Both core-to-router OSPF adjacencies reach FULL.

[ ] Core switches learn the OSPF default route.

[ ] Core-R1 reaches 203.0.113.2 and resolves its ARP entry.

[ ] PC reaches 203.0.113.2 and a PAT entry is visible.

Video walkthrough order

Topology > PC addressing > access VLANs > trunks > STP > SVIs > OSPF > DHCP relay > default route > PAT > successful ISP ping. For each topic, explain its purpose, show the relevant command and interpret the output.

Save & recovery

```
! Each configured router and switch
copy running-config startup-config
```

Save the Packet Tracer project with Ctrl+S to retain cloud mapping and device changes. Keep a known-good .pkt copy. If a change breaks the lab, undo the specific change using the captured baseline, or reopen the pre-change project copy.

Optional redundancy demonstration

Record the baseline, shut one identified redundant inter-switch uplink, verify STP recovery and client connectivity, then restore it with no shutdown. Do not shut the only working path. Do not claim gateway failover: HSRP is absent.

Source: user-supplied Week 11 MOP and topology screenshots, with the cloud DSL mapping correction and successful connectivity reported during troubleshooting. Device-specific port assignments must be verified against the running lab.